Lab 2.1 - Work Order Intake and Assignment

App: Microsoft 365 Copilot in **Excel**

Purpose: Use Microsoft 365 Copilot in Excel and Word to take two fictional drainage work orders from intake to the initial Pass/Fail decision, with a traceable status history.

Output: `Work-Order-Register.xlsx` and `Inspection-Plan.docx` covering WO-021 and WO-022.

Start with `Starter.xlsx`. The data is already built as a formatted Excel table. You do not import the CSV. The `.csv` files are a plain-text copy for the optional practice in Appendix A only.

Where this sits in Module 2

Lab	App	What you produce
2.1	**Excel**	**The work order register**
2.2	Word	The two inspection reports
2.3	Outlook	The approval request to the manager
2.4	PowerPoint	The weekly update to the supervisor

The scenario

A maintenance team inspects roadside drainage on the public road network. When a drain section is reported or falls due for inspection, a work order is raised, a worker attends the roadside with a graduated measuring rule, and the reading is compared against the specification maximum of **50 mm silt depth**.

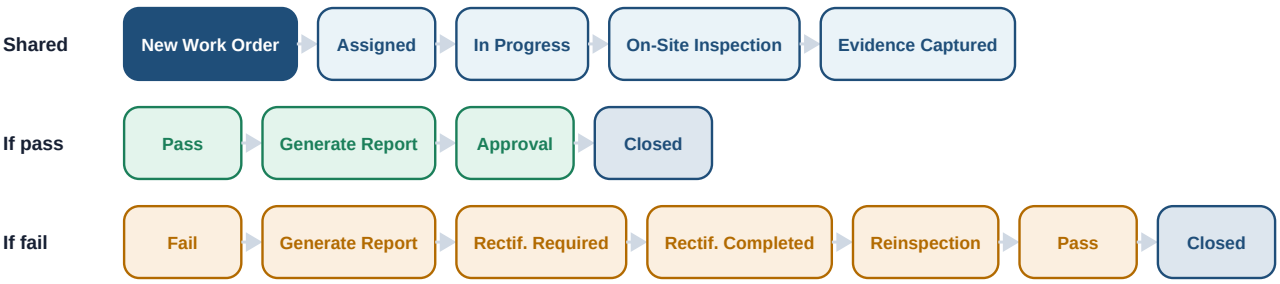
Two work orders are open this morning:

Work order	Asset	Location	Worker
WO-021	DR-17	Jalan Besar, drain section 17	Amira Tan
WO-022	DR-23	Clementi Avenue 2, drain section 23	Ben Lim

Roadside drainage matters for road safety, so an silted above the limit post cannot simply be closed. This lab takes both orders from intake to the moment the reading decides Pass or Fail. What happens after that decision is Lab 2.2.

The workflow you are following

Labs 2.1, 2.2 and 2.3 follow one work order journey. Every order starts on the shared path:



The green row is the pass branch; the amber row is the fail branch.

This lab covers the shared path only, up to and including the initial Pass/Fail. WO-021 will take the pass branch and WO-022 the fail branch, both in Lab 2.2.

The inspection criterion for both assets is **silt depth <= 50 mm**, from the fictional approved roadside drainage maintenance specification rev A. These are synthetic training records, not instructions for working on real road assets.

Two files, and which is which

You work with two kinds of file in Labs 2.1 to 2.3. Keep them straight and nothing else gets confusing.

File	What it is
`Starter.xlsx`	**Supplied data, one per lab.** Read-only starting material. Each lab folder has its own, holding that lab's tables. You open it to take data out of it, not to work in it.
`Work-Order-Register.xlsx`	**Your own working file.** You create it once, in Lab 2.1, by saving a copy of that lab's `Starter.xlsx`. It then follows you through Labs 2.2 and 2.3, growing as the work order progresses.

At the end of Lab 2.4 your register holds the whole journey: the intake and status history from 2.1, the inspections and rectification from 2.2, the approval log from 2.3, and the week's data from 2.4. That is the point of carrying one file forward — a reviewer can open it and see everything that happened, in order.

Files in this folder

File	Use
`Lab-Instructions.pdf`	This activity and every copy-ready Copilot prompt.
`Starter.xlsx`	**Open this one.** Formatted workbook, opens on a `Start_Here` sheet, with the `WorkOrders` table already built.
`mock-data.csv`	The same data as plain CSV. Not needed — for the optional import practice in Appendix A only.

Step 1 - Open the starter workbook

1. Open `Starter.xlsx`. If Excel shows a yellow **Protected View** bar, choose **Enable Editing**.

2. You land on the `Start_Here` sheet. It names the one thing to do first and lists what every other sheet is for. Read it, then move on.

Sheet	What it holds
`Start_Here`	Where to begin, and what each sheet is for.
`Work_Orders`	The two supplied work orders as the named table `WorkOrders`, plus the empty amber columns you will fill.
`Status_History`	An empty named table `StatusHistory` for the status rows you create.
`Workflow_Guide`	The workflow above, for reference while you work.
`Copilot_Prompt`	The same prompts printed in this PDF, ready to copy inside Excel.

3. On `Work_Orders`, confirm there are exactly two orders — WO-021 at DR-17 on Jalan Besar and WO-022 at DR-23 on Clementi Avenue 2.
4. Save the file as `Work-Order-Register.xlsx` with **File -> Save a Copy**. This copy is now *your* register — you work in it for the rest of this lab, and you reopen it in Labs 2.2 and 2.3. Leave `Starter.xlsx` untouched.

The data is already a formatted Excel table. You do not need to import anything, and you can ignore `mock-data.csv` unless your trainer asks for the import practice in Appendix A.

The amber columns `Current_Status`, `Status_Time_SGT`, `Decision`, `Worker_Review` and `Missing_Information` are already in place, ready for you to fill.

Step 2 - Select the named table

1. Click any cell inside the work order data.
2. Check the **Name Box**, at the left of the formula bar, shows `WorkOrders`.

Copilot in Excel works against the table you have selected. An unselected or wrongly named table is the most common reason a prompt comes back with nothing useful.

If the name is wrong or missing, rename it by hand: click inside the table, go to the **Table Design** tab, and type `WorkOrders` into the **Table Name** field on the far left. It must match exactly, with no space.

Copilot cannot do this for you. Renaming a table is a structural change to the workbook, not a data question, so it sits outside what Copilot in Excel will act on. Asking it to rename the table will not work, and the prompts in this lab refer to `WorkOrders` by name, so the name has to be right before you prompt.

Step 3 - Run the Excel prompt

1. With the `WorkOrders` table still selected, open **Copilot** from the Home tab.
2. Paste the prompt below, or copy it from the `Copilot_Prompt` sheet without leaving Excel.
3. Read the result before accepting it. Copilot proposes; you confirm.

Copy this prompt:

```
Use only the selected WorkOrders table on the Work_Orders sheet.

Create a second table called Status_History with the columns Work_Order,
Sequence, Status, Time_SGT, Actor, Source, Review_Note.

For each of the two work orders add these six rows in this order:
New Work Order, Assigned, In Progress, On-Site Inspection,
Evidence Captured, then the initial Pass or Fail.

Use only the supplied timestamps. Leave any time that was not supplied blank
and write NOT SUPPLIED in Review_Note.

In the WorkOrders table, fill Initial_Outcome as Pass when Initial_Reading is
less than or equal to 50 mm, otherwise Fail. Show the comparison you used.

Preserve every ID, name, asset, location, unit and criterion source exactly.
Do not invent evidence, approvals, permits, photographs or completed field work.
End with DRAFT FOR ENGINEER REVIEW.
```

Step 4 - Build the status history

Create the six pre-decision rows for **each** order on the `Status_History` sheet, in this order:

New Work Order, Assigned, In Progress, On-Site Inspection, Evidence Captured, then Pass or Fail.

Use one row per status change. Do not overwrite history with only the final status — the whole point of the register is that a reviewer can see the sequence. Leave any timestamp that was not supplied blank and write `NOT SUPPLIED` in `Review_Note`.

Step 5 - Check the decision by hand

Compare each initial reading against the supplied maximum yourself, before trusting any formula:

Order	Silt depth	Maximum	By hand	Outcome
WO-021	32 mm	15.0 mm	32 <= 50, inside by 18	**Pass**
WO-022	74 mm	15.0 mm	74 > 50, over by 24	**Fail**

A reading is not a cause analysis. WO-022 failing tells you nothing yet about why. Mark any missing site permit, calibration detail or worker sign-off as `NOT SUPPLIED`.

Step 6 - Draft the inspection plan in Word

1. Open a new Word document and open **Copilot**.
2. Attach `Work-Order-Register.xlsx`, or paste the work order table into the document.
3. Paste the prompt below.
4. Save as `Inspection-Plan.docx`, labelled DRAFT FOR ENGINEER REVIEW.

Copy this prompt:

```
Draft a two-work-order inspection plan using only the attached roadside drainage
work order table.

Include: scope; a table of work order, asset, location and assigned worker;
the 50 mm maximum silt depth and its fictional source; the measurement
and evidence-capture fields; the Pass/Fail decision rule; a missing-information
checklist; and blank worker and reviewer sign-off fields.

Reference WO-021 and WO-022 exactly as supplied. Mark anything not supplied as
NOT SUPPLIED. State DRAFT FOR ENGINEER REVIEW.
Do not claim that site access, calibration, a permit or an approval was verified.
```

Step 7 - Reconcile and save

Check that the Word plan and the Excel register carry the same work order IDs, assets, locations, workers, criterion and evidence IDs. Fix any mismatch in your own file rather than asking Copilot to paper over it.

Now save your work.

1. In Excel, press **Ctrl+S** (**Cmd+S** on Mac) to save `Work-Order-Register.xlsx`. Confirm the title bar shows that name and **not** `Starter.xlsx` — if it still says `Starter.xlsx`, you skipped the Save a Copy in Step 1, so do it now.
2. In Word, save `Inspection-Plan.docx` in the same folder.
3. Keep both files. **Lab 2.2 reopens** `Work-Order-Register.xlsx` **and adds to it**, so if you lose it you start the next lab without your status history.

Copy this prompt:

```
Compare the Word plan to the Excel table. List any mismatched IDs, values,
units or criterion source, and any invented fact. Do not silently repair them.
```

Pass conditions

- Both work orders have a traceable status history through the initial decision.
- Excel holds the supplied values and a Pass/Fail result you verified by hand.
- The Word plan states the criterion and its source and leaves sign-off fields blank.
- Both files stay marked as synthetic training work.
- The plan invents no real approvals, photographs, calibration certificates or safety permits.

Appendix A - Optional: import mock-data.csv yourself

You do not need this if you are using `Starter.xlsx`. Use it only for the import practice, or if the starter file is unavailable.

Do **not** simply double-click the CSV — that opens it as a plain CSV file with no table, and anything you add is lost when Excel closes it.

1. Open a **new blank Excel workbook**.

2. Go to the **Data** tab -> **Get Data** -> **From File** -> **From Text/CSV**.
3. Select `mock-data.csv` and choose **Import**.
4. In the preview window check that **Delimiter** is **Comma** and that the columns split correctly. Leave **Data Type Detection** as **Based on first 200 rows**.
5. Choose **Load To...** -> **Table** -> **New worksheet**, then **OK**. Excel creates a real table.
6. Click any cell in the new table. On the **Table Design** tab, set **Table Name** to `WorkOrders` — exactly that, no space. The prompts in this lab refer to the table by this name.
7. Rename the sheet tab to `Work_Orders`.
8. Add the columns `Current_Status`, `Status_Time_SGT`, `Decision`, `Worker_Review` and `Missing_Information` to the right of `Initial_Outcome`. Typing a heading in the first cell beside a table extends the table automatically.
9. Confirm `Initial_Reading` is right-aligned. If it is left-aligned Excel read it as text: select the column, then **Data -> Text to Columns -> Finish** to convert it to numbers.
10. Save as `Work-Order-Register.xlsx`, choosing **Excel Workbook (*.xlsx)** — not CSV.

All records in this lab are fictional training data. They describe no actual facility, person or organisation. Values are teaching assumptions, not approved design criteria. Do not use any output to operate, maintain or approve real equipment.